Q1 JEE Main 2020 - 7 Jan (Evening)

Let the tangents drawn from the origin to the circle, $x^2+y^2-8x-4y+16=0$ touch it at the point A and B. The $(AB)^2$ is equal to:

(A) $\frac{64}{5}$
(B) $\frac{24}{5}$
(C) $\frac{8}{5}$
(D) $\frac{8}{13}$

Q2 JEE Main 2020 - 8 Jan (Evening)

If a line, $y=mx+c$ is a tangent to the circle, $(x-3)^2+y^2=1$ and it is perpendicular to a line L_1, where L_1 is the tangent to the circle, $x^2+y^2=1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$;

(A) $c^2+6c+7=0$
(B) $c^2-6c+7=0$
(C) $c^2+6c-7=0$
(D) $c^2-6c-7=0$

Q3 JEE Main 2020 - 9 Jan (Morning)

If a circle touches y-axis at $(0, 4)$ and passes through $(2, 0)$ then which of the following can-not be the tangent to the circle

(A) $4x-3y+17=0$
(B) $3x-4y-24=0$
(C) $3x+4y-6=0$
(D) $4x+3y-8=0$

Q4 JEE Main 2020 - 9 Jan (Evening)

If the curves, $x^2-6x+y^2+8=0$ and $x^2-8y+y^2+16-k=0,\ (k>0)$ touch each other at a point, then the largest value of k is

Q5 JEE Main 2020 - 2 Sep (Morning)

The number of integral values of k for which the line, $3x+4y=k$ intersects the circle, $x^2+y^2-2x-4y+4=0$ at two distinct points is

Q6 JEE Main 2020 - 3 Sep (Morning)

The diameter of the circle, whose centre lies on the line $x+y=2$ in the first quadrant and which touches both the lines $x=3$ and $y=2$, is

Q7 JEE Main 2020 - 4 Sep (Evening)

The circle passing through the intersection of the circles, $x^2+y^2-6x=0$ and $x^2+y^2-4y=0$ having its centre on the line, $2x-3y+12=0$, also passes through the point:

(A) $(-3,6)$
(B) $(-1,3)$
(C) $(-3,1)$
(D) $(1,-3)$

Q8 JEE Main 2020 - 4 Sep (Evening)

Let PQ be a diameter of the circle $x^2+y^2=9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line, $x+y=2$ respectively, then the maximum value of $\alpha\beta$ is

Q9 JEE Main 2020 - 5 Sep (Evening)

If the length of the chord of the circle, $x^2+y^2=r^2(r>0)$ along the line, $y-2x=3$ is r then r^2 is equal to :

(A) $\frac{9}{5}$
(B) $\frac{24}{5}$
(C) $\frac{12}{5}$
(D) 12

Q10 JEE Main 2020 - 6 Sep (Evening)

The centre of the circle passing through the point (0,1) and touching the parabola $y=x^2$ at the point (2,4) is :

(A) $\left(\frac{-53}{10}, \frac{16}{5}\right)$
(B) $\left(\frac{6}{5}, \frac{53}{10}\right)$
(C) $\left(\frac{-16}{5}, \frac{53}{10}\right)$

(D) $\left(\frac{3}{10}, \frac{16}{5}\right)$

Answer Key

Q1 (A)	**Q2** (A)	**Q3** (D)	**Q4** (36)
Q5 (9)	**Q6** (3)	**Q7** (A)	**Q8** (7)
Q9 (C)	**Q10** (C)		

Q1

$L=\sqrt{S_1}=\sqrt{16}=4$
$R=\sqrt{16+4-16}=2$

Length of Chord of contact $=\dfrac{2LR}{\sqrt{L^2+R^2}}=\dfrac{2\times 4\times 2}{\sqrt{16+4}}=\dfrac{16}{\sqrt{20}}$

square of length of chord of contact $=\dfrac{64}{5}$

Q2

Slope of tangent to $x^2+y^2=1$ at $\left(\dfrac{1}{\sqrt{2}},\ \dfrac{1}{\sqrt{2}}\right)$

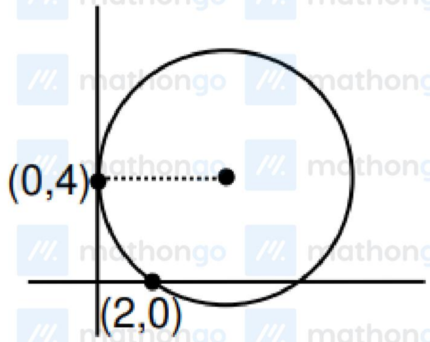
$x^2+y^2=1$
$2x+2yy'=0$
$y'=-\dfrac{x}{y}=-1$

$y=mx+c$ is tangent of $x^2+y^2=1$
so $m=1$
$y=x+c$
now distance of $(3,\ 0)$ from $y=x+c$ is

$\left|\dfrac{c+3}{\sqrt{2}}\right|=1$

$c^2+6c+9=2$
$c^2+6c+7=0$

Q3



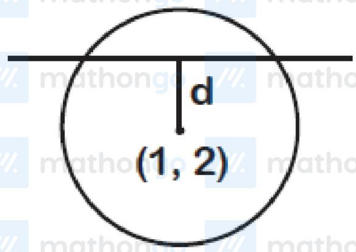
equation of family of circle
$(x-0)^2+(y-4)^2+\lambda x=0$
$\Rightarrow$ passes $(2,\ 0)$
$4+16+2\lambda=0 \quad \Rightarrow \quad \lambda=-10$
$x^2+y^2-10x-8y+16=0$
centre $(5,\ 4)\cdot R=\sqrt{25+16-16}=0$

Q4

Two circles touches each other if $C_1\ C_2\,|r_1 \pm r_2|$
Distance between $C_2\,(3,\ 0)$ and $C_1\,(0,\ 4)$ is either
$\sqrt{k}+1$ or $\left|\sqrt{k}-1\right| \quad (C_1\ C_2=5)$
$\Rightarrow \quad \sqrt{k}+1=5$ or $\left|\sqrt{k}-1\right|=5$
$\Rightarrow \quad k=16$ or $k=36$
$\Rightarrow \quad$ maximum value of k is 36

Q5

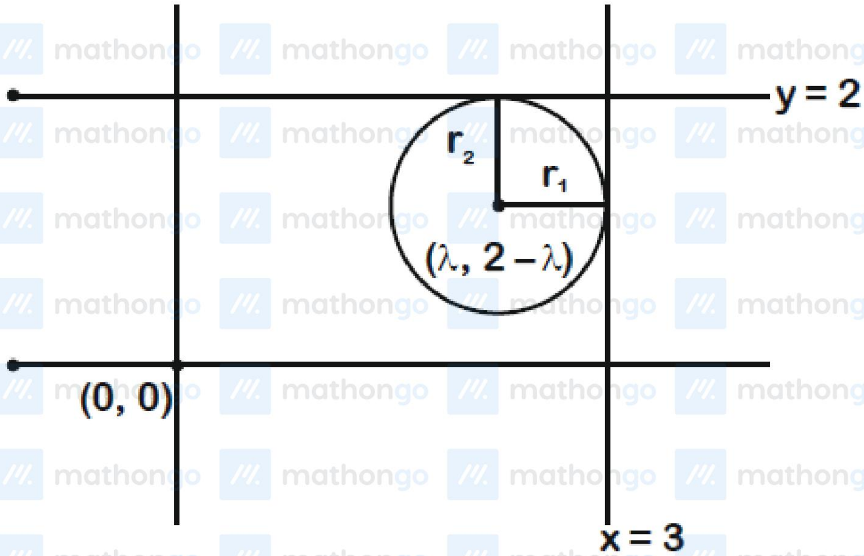
Given circle is $(x-1)^2+(y-2)^2=1$
$\Rightarrow \mathrm{d}<\mathrm{r}$
(where r is radius of circle)



$\Rightarrow \left|\frac{3(1)+4(2)-k}{\sqrt{3^2+4^2}}\right|<1$
$\Rightarrow |11-k|<5$
$\Rightarrow 6<k<16$

$\therefore \quad k = 7, 8, \ldots \ldots .15$
i.e. 9 values of k

Q6



As radius $= 3 - \lambda$
Also radius $= 2 - (2 - \lambda)$
$3 - \lambda = 2 - (2 - \lambda)$
$\Rightarrow \lambda = \frac{3}{2}$
$r = 3 - \frac{3}{2} = \frac{3}{2}$
Hence, diameter $= 3$

Q7

Let circle be $S_1 + \lambda S_2 = 0$
$x^2 + y^2 - 6x + \lambda\left(x^2 + y^2 - 4y\right) = 0$
$\Rightarrow (1+\lambda)x^2 + (1+\lambda)y^2 - 6x - 4\lambda y = 0$
Centre $\equiv \left(\frac{3}{1+\lambda}, \frac{2\lambda}{\lambda+1}\right)$
Centre lies on $2x - 3y + 12 = 0$ then
$\frac{6}{\lambda+1} - \frac{6\lambda}{\lambda+1} + 12 = 0$
$\Rightarrow \lambda = -3$
$C \equiv -2x^2 - 2y^2 - 6x + 12y = 0$

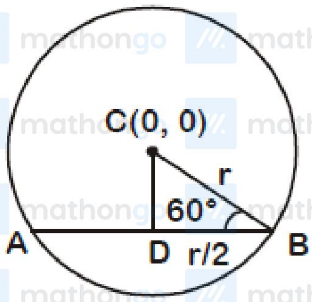
$x^2+y^2+3x-6y=0$
It passes through (-3,6)

Q8

Let $P(3\cos\theta, 3\sin\theta)$
$Q(-3\cos\theta, -3\sin\theta)$
$\alpha=\left|\frac{3\cos\theta+3\sin\theta-2}{\sqrt{2}}\right|$
$\beta=\left|\frac{-3\cos\theta-3\sin\theta-2|}{\sqrt{2}}\right|$
$\alpha\beta=\left|\frac{(3\cos\theta+3\sin\theta)^2-4}{2}\right.$
$=\left|\frac{5+9\sin 2\theta}{2}\right|$
$\left.\alpha\beta\right|_{\max}=\frac{5+9}{2}=7(\text{ when }\sin 2\theta=1)$

Q9

In right $\Delta\mathrm{CDB}$ -
$\sin 60^{\circ}=\frac{\mathrm{CD}}{r}$



$\Rightarrow \quad CD=r\times\frac{\sqrt{3}}{2}=\frac{\sqrt{3}r}{2}$
Now equation of AB is
$y-2x-3=0$
So $\frac{\sqrt{3}r}{2}=\frac{|0+0-3|}{\sqrt{5}}$
$\Rightarrow\frac{\sqrt{3}r}{2}=\frac{3}{\sqrt{5}}\Rightarrow r=\frac{2\sqrt{3}}{5}\Rightarrow r^2=\frac{12}{5}$

Q10

Circle passes through $A(0,1)$ and $B(2,4)$. So its centre is the point of intersection of perpendicular bisector of AB and normal to the parabola at (2,4) .
Perpendicular bisector of AB;
$y-\frac{5}{2}=-\frac{2}{3}(x-1) \Rightarrow 4x+6y=19$
Normal to the parabola at $(2,4)$,
$y-4=-\frac{1}{4}(x-2) \Rightarrow x+4y=18$
Centre of the circle is $\left(-\frac{16}{5}, \frac{53}{10}\right)$